

Chemical Water Sampling Report: River Towy (Golden Grove) SN 6159 21132

Sampler: Lee Nulty

Date of Sampling: 22nd May 2026

Time of Sampling: 11:00 AM

River Level (at Manorafon): 1.074 m

Water Temperature: 10.4°C

Water Condition and Observations: The water was slightly brown-tinted. The clean river bed and stones showed no signs of eutrophication, likely due to the recent rise in water levels. A gentle upstream breeze carried the distinctive smell of chemicals from the sewage outlet. Hatching invertebrates were observed, and several fish were seen taking insects from the surface above the outlet.

Executive Summary

This report evaluates the impact of a sewage discharge outlet on the water quality of the River Towy at Golden Grove. Samples were taken simultaneously upstream and downstream of the discharge point to isolate the environmental impact of the effluent. The data reveals a clear and measurable degradation of water quality downstream of the sewage outlet, characterised by sharp increases in key nutrient levels.

Data Comparison Table

Parameter	Upstream (SN 61671 21160)	Downstream (SN 6153 21101)	Percentage Change
Phosphate	0.03 mg/l	0.06 mg/l	+100% (Doubled)
Nitrate	0.5 mg/l	1.2 mg/l	+140% (More than doubled)
Ammonia	0.09 mg/l	0.13 mg/l	+44.4% Increase
pH	6.8	6.8	No change

Parameter Analysis and Ecological Meaning

1. Phosphate (PO_4^{3-})

- **The Change:** Levels doubled from 0.03 mg/l upstream to 0.06 mg/l downstream.
- **Meaning:** Phosphate is a primary limiting nutrient in freshwater ecosystems. The upstream level represents a relatively clean, oligotrophic to mesotrophic state. The jump to 0.06 mg/l downstream indicates a significant nutrient injection typical of domestic sewage (which contains detergents and organic waste). This increase accelerates eutrophication risk.

2. Nitrate (NO_3^-)

- **The Change:** Levels spiked from 0.5 mg/l to 1.2 mg/l downstream.
- **Meaning:** While 1.2 mg/l is well below dangerous toxicity thresholds for aquatic life, the 140% increase directly points to the sewage outlet as a significant source of nitrogenous pollution. In combination with elevated phosphates, this spike creates ideal conditions for excessive algal blooms. These blooms can choke out river oxygen supplies overnight.

3. Ammonia ($\text{NH}_3/\text{NH}_4^+$)

- **The Change:** Levels increased from 0.09 mg/l to 0.13 mg/l downstream.
- **Meaning:** Total ammonia levels above 0.1 mg/l often indicate recent, un-nitrified organic pollution (like raw or partially treated sewage). At a slightly acidic pH of 6.8 and a cool temperature of 10.4°C, most of this ammonia will exist in its non-toxic ionized form (ammonium). However, the upward trend represents a clear chemical stressor to sensitive fish species like salmon and trout, which use the Towy for spawning.

4. pH

- **The Change:** Stable at 6.8 at both test locations.
- **Meaning:** A pH of 6.8 is slightly acidic but entirely optimal for a Welsh river ecosystem. The sewage discharge did not possess enough acidity or alkalinity to overwhelm the natural buffering capacity of the river water at this flow level (1.074 m).

Conclusion

The sewage outlet is actively altering the chemistry of the River Towy. Even under cool conditions (10.4°C) and with a healthy river level providing dilution, the effluent causes a **100% increase in phosphates** and a **140% increase in nitrates**.

While the current concentrations do not represent immediate, acute toxicity, they indicate persistent chronic nutrient enrichment. If river levels drop during warmer summer months, these elevated nutrient inputs could trigger severe localised algal growth and oxygen depletion.